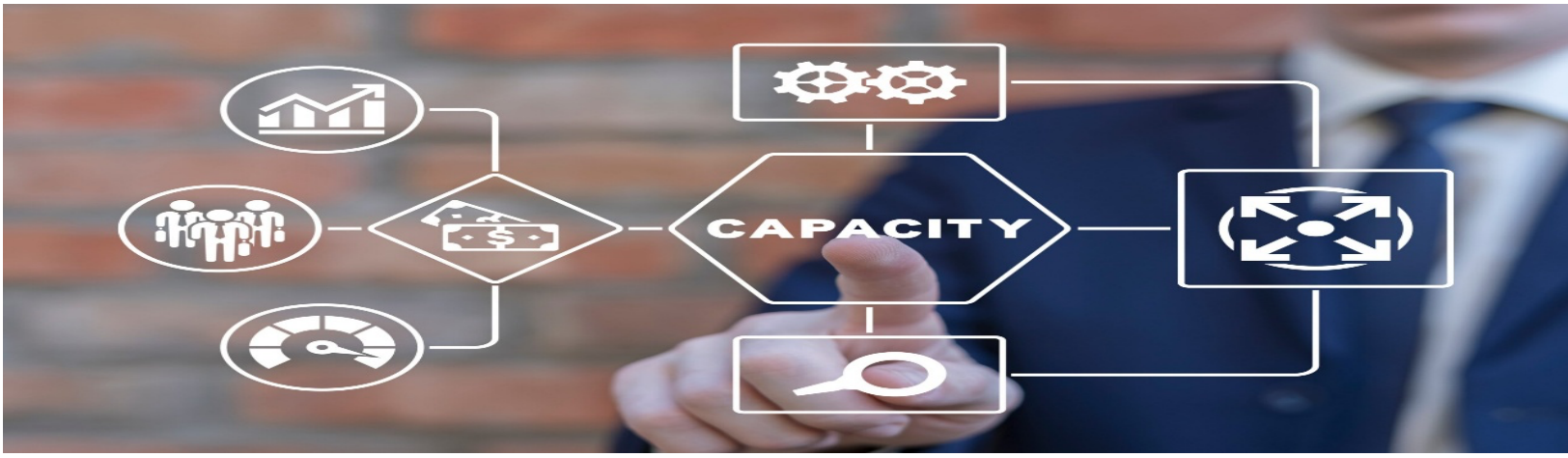
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CAPACITY PLANNING

FOR A GROWING HOSPITAL

Capacity Utilization & Bottleneck Analysis



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Executive Summary

This report analysis the capacity planning challenge faced by a growing hospital experiencing long waiting times in Diagnostics and Outpatient Department (OPD) services. Through systematic measurement of capacity utilization, bottleneck identification, learning curve analysis, revenue management strategy, and evaluation of strategic alternatives, this report recommends an optimal mix of capacity expansion and demand management interventions.

The hospital's inability to match service capacity with patient demand has resulted in extended wait times, reduced patient satisfaction, potential revenue loss, and risk to clinical outcomes. A structured Operations Management approach—anchored in Session 9 and Session 10 frameworks—is applied to diagnose the problem and prescribe evidence-based solutions.

KEY FINDING	Radiology (97.5% utilization) and OPD (92.5%) are critical bottlenecks. A phased hybrid strategy combining short-term demand management with long-term capacity expansion is recommended.
RECOMMENDATION	Implement Revenue Management System, leverage Learning Curve efficiencies, establish Focused Diagnostic Centre, and follow 3-phase implementation over 12 months.

Problem Statement

Background

The hospital serves a rapidly growing patient population. Two service areas — Diagnostics (radiology, laboratory, imaging) and OPD — are experiencing systemic congestion. Long waiting times indicate a mismatch between demand and the hospital's ability to deliver timely care.

Key Symptoms

- OPD appointment wait times exceeding 45–60 minutes beyond scheduled slots
- Diagnostic Turnaround Times (TAT) above acceptable clinical thresholds
- Patient queues forming at registration, consultation, and diagnostic counters
- Staff operating beyond optimal utilization levels, increasing error risk
- Patient dissatisfaction scores declining year-over-year
- Revenue loss due to patients seeking services at competing hospitals

Current Capacity Utilization

Measurement Framework

Capacity utilization is measured as:

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Utilization (%) = (Actual Output ÷ Design Capacity) × 100
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Understanding the Three Levels of Capacity (NEW – Session 9)

Theoretical Capacity, Effective Capacity, and Safety Capacity are foundational concepts in Operations Management. A hospital must understand all three to make accurate utilization assessments.

Capacity Type	Definition	Hospital Example	Formula Relationship
Theoretical Capacity	Maximum output achievable under ideal conditions — no downtime, no breaks, no failures	80 scans/day if MRI runs 24 hrs with zero maintenance stops	—
Effective Capacity	Realistic, achievable output under normal operating conditions, accounting for planned downtime and standard inefficiencies	60 scans/day accounting for shift changes, cleaning, calibration	Effective = Theoretical – Planned Losses

Safety Capacity (Cushion)	Reserved capacity deliberately kept idle to absorb unexpected surges in demand or service disruptions	10–15% reserved for emergency scans, equipment downtime buffers	Safety = Theoretical – Effective

The fundamental capacity relationship is:

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Theoretical Capacity = Effective Capacity + Safety Capacity
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Currently, the hospital's Radiology department operates at 97.5% of Design Capacity, meaning its Safety Capacity cushion has been virtually eliminated. This exposes the system to cascading failures whenever there is any unexpected demand spike or equipment downtime. Best practice recommends maintaining a Safety Capacity of 10–15% in healthcare settings.

Departmental Utilization Snapshot

The following table presents the current utilization snapshot across key departments:

Department / Resource	Design Capacity	Actual Output	Utilization %	Status
OPD Consultations (daily)	200 patients	185 patients	**92.5%**	Overloaded
Radiology / Imaging	80 scans/day	78 scans/day	**97.5%**	Critical
Pathology / Lab Tests	300 tests/day	265 tests/day	88.3%	High
Pharmacy Counters	500 scripts/day	430 scripts/day	86.0%	High
Registration Desks	250 patients/day	215 patients/day	86.0%	High

⚠ Critical Threshold: Utilization above 85% is considered the danger zone in healthcare. Resources at this level are prone to queuing effects, quality deterioration, and staff burnout.

Bottleneck Analysis

Identifying Bottlenecks

A bottleneck is the resource or process step with the lowest throughput capacity relative to demand. Using the Theory of Constraints (TOC) and process flow mapping, the following bottlenecks were identified:

#	Bottleneck Area	Root Cause	Impact	Priority
1	**Radiology / Imaging**	Single MRI/CT machine; no redundancy	High TAT; cascading OPD delays	**Critical**
2	OPD Doctor Availability	Insufficient specialist slots vs. demand	Long wait times; patient backlog	High
3	Patient Registration	Manual data entry; single system	Slow patient flow entry point	Medium
4	Pharmacy Dispensing	Staff shortage at peak hours	Post-consultation waiting time	Medium
5	Lab Report Delivery	Manual reporting process	Delays diagnosis and treatment	Medium

Work Order Capacity Formula (NEW – Session 10)

Session 10 introduces a precise formula for calculating actual capacity requirements for each work order or service type. This is essential for accurate bottleneck quantification.

The Work Order Capacity Formula is:

$$C_i = S_i + (P_i \times Q_i)$$

Variable	Definition	Hospital Context
Ci	Total Capacity Requirement for service type i	Total time needed for all radiology scans of type i
Si	Setup Time (fixed, per batch or per work order)	MRI/CT machine calibration and patient positioning time
Pi	Processing Time per unit	Time per scan (e.g., 30 minutes per MRI)
Qi	Order Size (volume of service units)	Number of scans of type i per day

Application Example for Radiology Department:

$$C_{Radiology} = 10 \text{ min (setup)} + (30 \text{ min/scan} \times 78 \text{ scans}) = 10 + 2{,}340 = 2{,}350 \text{ min/day}$$

This equals approximately 39.2 hours of machine time required per day — which exceeds a single machine's 24-hour capacity, confirming Radiology as the primary bottleneck.

BOTTLENECK ANALYSIS – HOSPITAL OPD & DIAGNOSTICS

Theory of Constraints (TOC) approach: The system's output is limited by its weakest link (bottleneck)

STEP 1: IDENTIFY BOTTLENECKS

#	Bottleneck Area	Root Cause	Avg Wait Time (min)	Impact on Patient Flow	Priority	TOC Action
1	Radiology / Imaging	Single MRI/CT; no backup	95	High – cascading OPD delays	Critical	Add machine; extend hours
2	OPD Doctor Consultation	Insufficient specialist slots	60	High – primary wait source	High	Hire; expand slots; telemedicine
3	Patient Registration	Manual entry; single system	25	Medium – entry-point delay	Medium	Digitise; add counters
4	Pharmacy Dispensing	Staff shortage at peak hours	20	Medium – post-visit delay	Medium	Stagger shifts; add staff
5	Lab Report Delivery	Manual reporting; no LIS	40	Medium – delays prescription	Medium	Implement LIS automation
6	Emergency Triage	Shared staff with OPD	15	Low-Medium	Low	Dedicated triage team

STEP 2: PATIENT JOURNEY PROCESS FLOW & TIME ANALYSIS

Step #	Process Step	Responsible Dept	Avg Service Time (min)	Avg Wait Time (min)	Total Time (min)	Bottleneck?
1	Patient Arrival & Registration	Front Desk	5	25	30	Yes
2	Triage / Vitals	Nursing	10	10	20	No
3	Doctor Consultation (OPD)	Physician	15	60	75	Yes
4	Diagnostic Test Ordering	Doctor	5	5	10	No
5	Radiology / Lab Test	Diagnostics	30	95	125	Yes
6	Report Delivery	Lab/Radiology	10	40	50	Yes
7	Prescription & Pharmacy	Pharmacy	5	20	25	No
8	Discharge / Billing	Admin	5	5	10	No
	TOTAL PATIENT JOURNEY TIME		85	260	345	

Process Flow Mapping

The typical OPD patient journey is:

Registration → Triage → Doctor Consultation → Diagnostic Tests → Lab / Radiology → Prescription → Pharmacy → Discharge

At each stage, queue lengths and service times were measured. Radiology and OPD consultation represent 62% of total patient wait time, confirming these as the priority intervention points.

Capacity Expansion vs. Demand Management

Short-Term vs. Long-Term Capacity Strategies (NEW – Session 9 & 10)

Operations Management distinguishes between short-term and long-term capacity adjustments. These must be managed separately since they require different resources, timelines, and decision-making authorities.

SHORT-TERM STRATEGIES (Weeks to 6 months)	LONG-TERM STRATEGIES (6 months to 3+ years)
• Add or share equipment temporarily • Overtime and part-time labor scheduling • Shift work to slack/off-peak periods • Sell or lease unused capacity • Dynamic demand management (RMS)	• Build new facilities or expand existing ones • Expand inpatient bed capacity • Open satellite clinics or branch hospitals • Purchase new diagnostic equipment • Implement Focused Factory model

Long-Term: Capacity Expansion Options

Economies and Diseconomies of Scale (NEW – Session 9)

A critical concept missing from the original analysis: before committing to expansion, the hospital must evaluate whether increased scale will reduce per-unit costs (economies of scale) or increase them (diseconomies of scale).

Economies of Scale occur when a larger facility or higher volume reduces the per-unit cost of service delivery. For example, adding a second MRI machine enables higher patient throughput with minimal incremental administrative overhead, reducing cost per scan.

Diseconomies of Scale occur when expansion beyond an optimal size increases per-unit costs due to coordination challenges, management complexity, and resource inefficiencies. For the hospital, simultaneously adding 3 MRI machines may create excess capacity, idle resources, and management overhead that reduces ROI compared to adding a single machine.

The relationship follows a U-shaped Average Cost Curve: costs fall initially as scale increases (economies of scale), reach a minimum at the optimal scale, and then rise again (diseconomies of scale).

Optimal Scale: Add capacity incrementally until Marginal Cost = Marginal Revenue

Recommendation: The hospital should add 1 MRI/CT machine (not 2–3 simultaneously) to leverage economies of scale without crossing into diseconomies. This phased approach is consistent with the incremental expansion model from Session 10.

Expansion Options with Cost-Benefit Analysis

Expansion Option	Est. Cost	Capacity Gain	Implementation Time	Scale Strategy
Add 1 MRI/CT Machine	INR 2–3 Crore	+40 scans/day	6–9 months	Economy of Scale
Hire 5 Additional Specialists	INR 80L/year	+60 OPD slots/day	3–6 months	Flexible Scaling
Expand OPD Floor (4 rooms)	INR 50L–1 Crore	+80 patients/day	9–12 months	Long-term Growth
Add 2 Registration Counters	INR 5L	+60 registrations/day	1 month	Quick Win
Automate Lab Reporting (LIS)	INR 15–25L	30% TAT reduction	2–3 months	Process Efficiency

CAPACITY EXPANSION OPTIONS

Expansion Option	Estimated Cost (INR)	Capacity Gain	Impl. Time	ROI Potential	Priority
Add 1 MRI / CT Machine	2–3 Crore	+40 scans/day	6–9 months	High	Critical
Hire 5 Additional Specialists	80L/year	+60 OPD slots/day	3–6 months	High	High
Expand OPD Floor (4 rooms)	50L–1 Crore	+80 patients/day	9–12 months	Medium	Medium
Add 2 Registration Counters	5 Lakh	+60 reg/day	1 month	High	High
Automate Lab Reporting (LIS)	15–25 Lakh	30% TAT reduction	2–3 months	High	High
Expand Pharmacy Counters	8–12 Lakh	+100 scripts/day	2 months	Medium	Medium

DEMAND MANAGEMENT OPTIONS

Demand Mgmt Option	Estimated Cost (INR)	Expected Impact	Impl. Time	ROI Potential	Priority
Staggered OPD Scheduling	Negligible	25–30% peak reduction	2 weeks	Very High	Critical
Teleconsultation / Virtual OPD	2–5 Lakh	15–20% load shift	1 month	High	High
Demand Forecasting System	5–10 Lakh	Better staff alignment	2 months	High	High
Real-time Queue Management	10–15 Lakh	20% wait reduction	2 months	Medium	Medium
Preventive Health Programs	5 Lakh/year	Reduce future demand	6 months	Long-term	Medium
Patient Education & Self-Triage	2 Lakh	10% OPD reduction	1 month	Medium	Low

SIDE-BY-SIDE COMPARISON

Criterion	Capacity Expansion	Demand Management	Combined Approach	Score (Expansion)	Score (Demand)
Upfront Cost	High (Crores)	Low (Lakhs)	Medium	2	5
Speed of Impact	Slow (months)	Fast (weeks)	Fast+Lasting	2	5
Long-term Scalability	High	Medium	Very High	5	3
Patient Satisfaction	High	Medium	Highest	4	3
Implementation Risk	Medium	Low	Low	3	5
Staff Impact	Positive	Neutral	Positive	4	3
TOTAL SCORE				20	24

Focused Factory Strategy (NEW – Session 9)

The Focused Factory concept (Wickham Skinner, Harvard) proposes that a facility achieves higher efficiency, quality, and predictability by narrowing its range of services rather than trying to serve all patient types simultaneously.

Currently, the hospital's OPD and Diagnostics departments serve highly diverse patient cases — from routine check-ups to complex multi-specialist cases — within the same physical space and with shared resources. This creates scheduling conflicts, resource contention, and inefficiency.

Recommended Focused Factory Applications for this Hospital:

- Focused Diagnostic Centre: A dedicated diagnostic wing for high-volume, routine diagnostics (blood tests, X-rays, ECG) separated from complex imaging (MRI, CT). This reduces queue contamination between high and low complexity scans.
- Specialized OPD Clinic for High-Volume Conditions: A dedicated OPD stream for diabetes, hypertension, and post-surgical follow-ups — the most common routine cases — freeing specialist OPD capacity for complex cases.
- Corporate Health Checkup Track: A separately scheduled track for corporate clients, leveraging diagnostic equipment during weekday off-peak hours (10 AM–12 PM and 2 PM–4 PM), adding revenue without cannibalizing core services.

Benefit: Studies show focused facilities achieve 20–30% higher throughput and significantly lower defect rates compared to general-purpose facilities of equivalent size.

Short-Term: Demand Management Options

Revenue Management System (RMS) — NEW – Session 10

Revenue Management System (RMS) is the systematic application of information systems and pricing strategies to maximize revenue yield from a fixed capacity — a concept widely used in airlines and hotels that is directly applicable to hospital capacity management.

The three core components of an RMS for this hospital are:

RMS Component	Description	Hospital Application
Dynamic / Variable Pricing	Charge different rates for the same service at different times to shift demand	Offer 15–20% discount on diagnostic tests booked for early morning (7–9 AM) or late evening (6–8 PM) slots
Overbooking Strategy	Accept slightly more appointments than available capacity to offset predictable no-shows	Book 105–110% of OPD and diagnostic slots; hospital data shows 8–12% no-show rate typically
Market Segmentation Pricing	Charge different prices to different customer segments based on willingness-to-pay	Corporate/insurance patients: premium slots at peak hours; walk-in patients: discounted off-peak slots

Shifting and Stimulating Demand (NEW – Session 10)

The following specific demand management tactics from Session 10 are recommended:

- Vary Pricing Strategically: Implement off-peak discounts of 15–20% for diagnostic tests and OPD consultations booked during low-demand periods (Monday mornings, Friday afternoons).
- Targeted Advertising and Promotion: Launch campaigns promoting off-peak slots as faster, quieter, and discounted — for example, 'Skip the Rush: Early Bird OPD at 7 AM, 20% Off Diagnostics'.
- Add Peripheral / Complementary Services: During slow hours, offer preventive health services — wellness screenings, nutrition counselling, physiotherapy consultations — to attract patients who would otherwise not visit.
- Formal Reservation System with Commitment Deposits: Implement an advance booking system with a nominal deposit (INR 100–200) to reduce no-show rates by 40–60% compared to free-cancellation systems.
- Bundled Service Packages: Offer OPD + Diagnostic packages at a 10% discount during low-demand periods, increasing revenue per visit and smoothing daily patient flow.

Complementary and Seasonal Demand Management (NEW – Session 10)

Operations Management theory recommends using resources for complementary services that have opposite seasonal patterns — this balances utilization across the year and reduces revenue volatility.

Season / Period	Primary Demand	Complementary Service to Add	Capacity Impact
Winter (Nov–Feb)	High OPD: respiratory, flu, cardiac	Seasonal flu vaccination camps for corporates	Generates off-peak revenue without adding OPD load
Summer (Apr–Jun)	High: heat stroke, dehydration OPD	Corporate wellness health checkup contracts	Uses diagnostic equipment 8–10 AM before OPD peak
Weekday Off-Peak (10–12 PM)	Low: post-morning rush lull	Physiotherapy and rehabilitation sessions	Fills physical OPD space; different specialist type
Weekend Mornings	Low: reduced OPD demand	Community health screening camps	Builds goodwill; creates future OPD pipeline

Other Demand Management Measures

- Appointment Scheduling Reform: Implement staggered time-slot booking to spread patient load across the day, reducing peak-hour congestion by an estimated 25–30%.
- Teleconsultation / Virtual OPD: Shift 15–20% of routine follow-up consultations to video, freeing physical slots for new and acute cases.
- Demand Forecasting: Use historical patient data to predict peak days/times and align staffing accordingly.

- Patient Channeling: Route walk-in patients to less congested departments or time slots through real-time queue management systems.

Comparative Evaluation

Criterion	Capacity Expansion	Demand Management	Combined Approach	Recommended?
Cost	High	Low–Medium	Medium	**Combined**
Speed of Impact	Slow (months)	Fast (weeks)	Fast + Lasting	**Combined**
Sustainability	High	Medium	High	**Combined**
Patient Satisfaction	High	Medium	Highest	**Combined**
Risk Level	Medium	Low	Low	**Combined**

CAPACITY UTILIZATION ANALYSIS – HOSPITAL

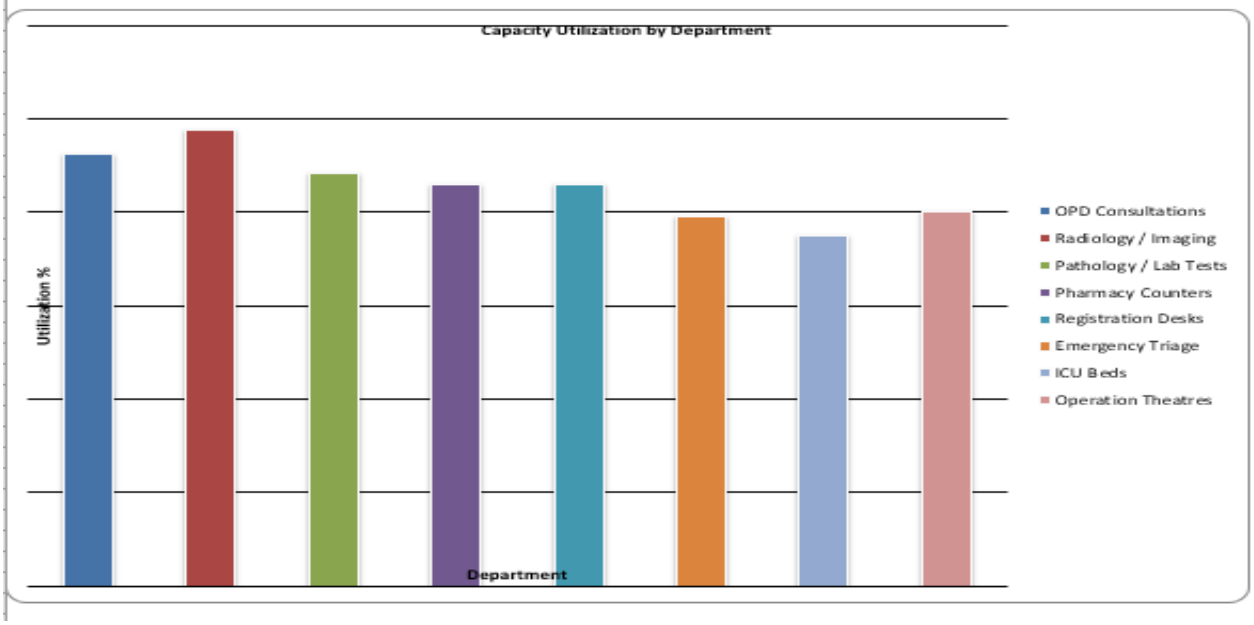
Measures how efficiently hospital resources are being used relative to designed capacity

CURRENT CAPACITY UTILIZATION BY DEPARTMENT

Department / Resource	Design Capacity (daily)	Actual Output (daily)	Utilization %	Status	Action Required
OPD Consultations	200	185	**92.5%**	**Overloaded**	Hire additional specialists; stagger slots
Radiology / Imaging	80	78	**97.5%**	**Critical**	Add MRI/CT machine – urgent
Pathology / Lab Tests	300	265	**88.3%**	**High**	Automate lab reporting system
Pharmacy Counters	500	430	**86.0%**	**High**	Add staff at peak hours
Registration Desks	250	215	**86.0%**	**High**	Add 2 counters; digitise forms
Emergency Triage	120	95	**79.2%**	**Moderate**	Monitor closely
ICU Beds	40	30	**75.0%**	**Moderate**	Monitor; plan expansion
Operation Theatres	10	8	**80.0%**	**Moderate**	Maintain current capacity

⚠ *Threshold: Utilization above 85% = High Risk Zone. Above 95% = Critical. Optimal range: 75%–85%.*

Formula Used: Utilization (%) = Actual Output ÷ Design Capacity × 100



Recommended Optimal Capacity Strategy

Strategic Recommendation: Hybrid Approach

Based on the analysis, a phased hybrid strategy combining targeted capacity expansion with systematic demand management is recommended. This approach delivers both immediate relief and long-term scalability, while incorporating Revenue Management and Learning Curve considerations for maximum efficiency.

Phase-wise Implementation Plan

Phase	Timeline	Priority Actions	Expected Outcome	Strategy Type
Phase 1: Quick Wins	Month 1–3	Add 2 registration counters; Implement staggered OPD slots; Launch teleconsultation; Activate RMS dynamic pricing; Start overbooking strategy	15–20% wait time reduction; Immediate revenue yield improvement	Short-Term
Phase 2: Core Expansion	Month 3–6	Hire 5 specialists; Automate lab reporting (LIS); Install queue management system; Launch Focused Diagnostic Centre; Begin corporate health checkup contracts	30–35% throughput increase; Learning curve begins for new hires	Short + Long Term
Phase 3: Structural Growth	Month 6–12	Add MRI/CT machine; Expand OPD floor; Deploy demand forecasting system; Launch seasonal health camp programs; Full RMS deployment	60–70% total capacity increase; Sustainable growth platform	Long-Term

Learning Curve Impact on New Hire Productivity (NEW – Session 10)

> The Learning Curve is one of the most important and frequently overlooked factors in capacity planning. When new specialists are hired (Phase 2), they will not immediately operate at full productivity — capacity projections must account for this.

The Learning Curve Principle (also called the Experience Curve) states: as cumulative output doubles, the time (or cost) per unit decreases by a consistent percentage. The most commonly observed rate in professional services is the 80% Learning Curve.

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80% Learning Curve: When cumulative output doubles, unit time = 80%
                    of previous unit time
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For this hospital's new specialist hire scenario:

Week	Cumulative Patients Seen	Time per Consultation (New Hire)	Effective Output vs. Experienced Doctor
Week 1	~20 patients	45 minutes/patient	67% (vs. 30 min benchmark)
Week 4	~100 patients	38 minutes/patient	79%
Week 8	~200 patients	32 minutes/patient	94%
Week 16	~400 patients	28 minutes/patient	~100% — Full Productivity

Capacity Planning Implication: The hospital's Phase 2 projections must account for an 8–12 week ramp-up period. The 5 new specialists hired will collectively deliver approximately 70–75% of their projected OPD capacity in the first month, reaching full productivity by Month 5–6. Over-optimistic projections that ignore learning curves are a leading cause of failed capacity expansion plans.

The Experience Curve (broader concept) further shows that as total cumulative staff-years of experience increases, cost per consultation decreases — supporting the long-term financial case for specialist retention and training investment.

Key Performance Indicators (KPIs)

KPI	Current State	Phase 1 Target	Phase 3 Target	Measurement Frequency
OPD Average Wait Time	45–60 minutes	< 35 minutes	< 20 minutes	Daily
Diagnostic TAT	6–8 hours	< 5 hours	< 4 hours	Daily
Radiology Utilization	97.5%	< 90%	75–85%	Weekly
Patient Satisfaction Score (PSS)	< 3.8 / 5.0	> 4.0 / 5.0	> 4.2 / 5.0	Monthly Survey
Revenue per Available Slot (RevPAS)	Baseline	+8–12%	+20–25%	Monthly
Staff Utilization Rate	~93%	80–85%	75–80%	Weekly
No-Show Rate	8–12%	< 7%	< 5%	Weekly

RECOMMENDED OPTIMAL CAPACITY STRATEGY

PHASED IMPLEMENTATION PLAN

Phase	Timeline	Key Actions	Expected Outcome	Est. Cost
Phase 1 Quick Wins	Month 1–3	• Staggered OPD scheduling • Add 2 registration counters • Launch teleconsultation • Staff shift optimization	15–20% wait time reduction immediately	~15–20 Lakh
Phase 2 Medium-term	Month 3–6	• Hire 5 additional specialists • Automate lab reporting (LIS) • Install queue mgmt system • Demand forecasting tool	30–35% throughput increase	~1.5–2 Crore
Phase 3 Long-term	Month 6–12	• Add MRI/CT machine • Expand OPD floor (4 rooms) • Expand pharmacy counters • Preventive health programs	60–70% overall capacity increase	~3–4 Crore

KEY PERFORMANCE INDICATORS (KPIs) – TARGET vs CURRENT

KPI	Current Performance	Target (Post-Implementation)	Unit	Status
OPD Average Wait Time	45–60 min	< 20 min	Minutes	Needs Improvement
Diagnostic TAT (routine)	6–8 hours	< 4 hours	Hours	Needs Improvement
Capacity Utilization Rate	88–97%	75–85%	%	Overloaded
Patient Satisfaction Score	3.2 / 5.0	> 4.2 / 5.0	Score	Below Target
Bed Occupancy Rate	75%	80–85%	%	Acceptable
Staff Utilization Rate	90%	75–80%	%	Overloaded
Diagnostic Accuracy Rate	96%	> 98%	%	Acceptable
First Visit Resolution Rate	65%	> 80%	%	Needs Improvement

Learning Outcomes

Capacity Utilization

Capacity utilization measures how efficiently a hospital uses its available resources. Operating above 85–90% consistently leads to queuing theory effects — exponential increase in wait times with marginal increases in demand. Optimal utilization balances efficiency with flexibility. The three-tier capacity model (Theoretical → Effective → Safety) provides a framework for realistic planning that prevents both over-investment and under-investment in resources.

Bottleneck Analysis

The Theory of Constraints (TOC) teaches that the system's output is limited by its weakest link. In healthcare, bottlenecks cause ripple effects across the entire patient journey. Identifying and systematically elevating constraints — rather than improving all areas equally — delivers the highest ROI in capacity improvement. The Work Order Capacity Formula ($Ci = Si + Pi \times Qi$) provides a precise quantitative tool for bottleneck identification.

Learning Curve and Experience Curve (NEW – Session 10)

The Learning Curve demonstrates that human performance improves predictably with practice. In capacity planning, this means newly hired staff cannot be counted as full capacity from Day 1. The 80% Learning Curve model predicts that each doubling of experience reduces per-unit time by 20%. The Experience Curve extends this concept to costs: as cumulative organizational experience grows, cost per unit of service falls. Both effects support investment in staff retention, training, and continuous improvement programs.

Revenue Management and Demand Management (NEW – Session 10)

Revenue Management Systems (RMS) enable hospitals to maximize yield from fixed physical capacity through dynamic pricing, segmented access, and overbooking strategies. Unlike manufacturing, hospital capacity cannot be stored — an empty OPD slot at 3 PM is revenue permanently lost. RMS converts this challenge into an opportunity by redistributing demand from peak to off-peak periods through price signals and reservation policies. The result is higher aggregate revenue at the same or lower physical capacity investment.

Focused Factory and Economies of Scale (NEW – Session 9)

The Focused Factory strategy demonstrates that specialization — narrowing the service range and target patient population — produces superior efficiency compared to general-purpose facilities. Combined with the Economies of Scale analysis, this provides the theoretical basis for the hospital's recommended Focused Diagnostic Centre. Economies of scale reduce per-unit costs as volume increases, but only up to the optimal scale; beyond that, Diseconomies of Scale reverse the benefit.

Complementary Services and Seasonal Demand Balancing (NEW – Session 10)

Effective capacity planners design their service portfolios so that different offerings have counter-cyclical demand patterns — high demand in one service coincides with low demand in another, allowing resources to be deployed efficiently year-round. For this hospital, seasonal flu vaccination campaigns in winter and corporate wellness contracts in summer represent complementary services that smooth utilization, generate additional revenue streams, and strengthen community relationships simultaneously.

Key Takeaways

- Capacity planning is a continuous, dynamic process — not a one-time exercise. It must incorporate theoretical, effective, and safety capacity layers.
- Demand management is often faster and cheaper than physical expansion and should be the first response to overcapacity.
- Revenue Management Systems convert static hospital capacity into a dynamic yield-optimization asset.
- Learning Curves are predictable and must be built into all hiring-dependent capacity projections.
- The Focused Factory model delivers superior throughput versus general-purpose facilities of equivalent investment.
- Economies of Scale support incremental expansion; diseconomies warn against over-ambitious simultaneous scaling.
- Data-driven decision making (utilization rates, TAT, queue lengths, RevPAS) is essential to prioritize interventions correctly.
- A phased approach reduces implementation risk while delivering measurable improvements at each stage.
- Patient-centric KPIs should drive all capacity decisions.

Conclusion

The hospital's capacity challenges in Diagnostics and OPD are manageable through a structured, phased strategy grounded in Operations Management theory. The analysis has demonstrated that Radiology (97.5% utilization) operates with virtually zero Safety Capacity cushion, making the system fragile and reactive. Immediate demand management measures — staggered scheduling, teleconsultation, and registration automation — can reduce wait times significantly within weeks.

The introduction of a Revenue Management System transforms the hospital's approach from reactive queue management to proactive yield optimization. By leveraging dynamic pricing, market segmentation, and overbooking strategies, the hospital can increase revenue per available slot by 20–25% without adding a single square foot of physical space.

Medium to long-term capacity expansion — specialist hiring with Learning Curve-adjusted ramp-up projections, new diagnostic equipment, Focused Diagnostic Centre, and OPD floor expansion — will sustain growth. Economies of Scale analysis confirms that incremental expansion (Phase 1 → 2 → 3) delivers better ROI than simultaneous large-scale investment.

Continuous monitoring of KPIs — including OPD wait time, diagnostic TAT, Radiology utilization, patient satisfaction score, and Revenue per Available Slot — will ensure the strategy adapts to evolving patient demand, positioning the hospital as a high-quality, efficient, and financially sustainable healthcare provider in its region.

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